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n = 4 ; f orbital $l = 3 \Rightarrow m_l = -3, -2, -1, 0, +1, +2$

Out of two electrons, one electron occupies 4f orbital with $m_l = -3$ and another electron occupies 4f orbital with $m_l = -2$.

∴ All the four quantum numbers for the two electrons are

Electron	n	l	m_l	m_s
$1e^-$	4	3	–3	+1/2
$2e^-$	4	3	–2	+1/2

8.

Electronic configuration of Fe^{3+} : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^5$

Electronic configuration of Ni^{2+} : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^8$

Fe^{3+} has stable $3d^5$ half filled configuration.

EVALUATION

I Choose the best answer:

1. (c) 30
2. (c) 4.42×10^{-18} J
3. (b) $\lambda_1 = 2\lambda_2$
4. (d) Stark effect
5. (b)
6. (d) n=6 to n=5
7. (a)
8. (c) d_{z2}, $d_{x^2-y^2}$
9. (b) Spin quantum number
10. (b) $[Xe]4f^7\ 6s^2$; $[Xe]4f^7\ 5d^1\ 6s^2$; $[Xe]4f^9\ 6s^2$
11. (c) 4l+2
12. (d) $\sqrt{6}\ h/2\pi$
13. (c) 2
14. (c)
15. (a) 9
16. (a) ns→(n-2)f→(n-1)d→np
17. (b) (ii),(iv) & (v)
18. (b) 17
19. (a) Zero
20. (c) $1/2m\ \sqrt{h/\pi}$
21. (c) 6.6×10^{-31} cm
22. (d) 0.4
23. (d) -9E
24. (a) $\hat{H}\psi = E\psi$
25. (d) $\Delta E.\Delta X \geq h/4\pi$